NOTES ON ACCURATE BOTTOM BC ON THE BUOYANCY EQUATION

GPT 5.5 WITH PROMPTING FROM TWNH

1. Boundary-reduced solution using a Dirichlet-to-Neumann map

Consider the steady buoyancy problem in the domain

$$\Omega = \{(x, y, z) : -H(x, y) \le z \le 0\}.$$

For simplicity, take the interior equation to contain only vertical diffusion,

$$\varepsilon^2 \partial_z \left(\kappa \partial_z b\right) + \gamma(B - b) = 0, \qquad -H(x, y) < z < 0,$$

with inhomogeneous surface Dirichlet condition

$$b(x, y, 0) = b_s(x, y),$$

and no-flux condition on the sloping bottom

$$\left[b_z + \delta^2 \nabla_h b \cdot \nabla_h H\right]_{z=-H(x,y)} = 0.$$

Here

$$\nabla_h = (\partial_x, \partial_y).$$

The difficulty is that the bottom boundary condition couples neighboring water columns through the term

$$\nabla_h b \cdot \nabla_h H.$$

A convenient way to isolate this coupling is to reduce the problem to an equation posed only on the bottom boundary.

Define the bottom trace

$$\beta(x, y) = b(x, y, -H(x, y)).$$

The horizontal gradient of the bottom trace is not the same as the horizontal gradient of b evaluated at fixed z. By the chain rule,

$$\nabla_h \beta = \nabla_h b|_{z=-H} + b_z|_{z=-H} \nabla_h(-H).$$

Since

$$\nabla_h(-H) = -\nabla_h H,$$

we have

$$\nabla_h \beta = \nabla_h b|_{z=-H} - b_z|_{z=-H} \nabla_h H.$$

Therefore,

$$\nabla_h b|_{z=-H} = \nabla_h \beta + b_z|_{z=-H} \nabla_h H.$$

Date: 26Jun26.

Substituting this into the bottom no-flux condition gives

$$b_z + \delta^2 \left(\nabla_h \beta + b_z \nabla_h H\right) \cdot \nabla_h H = 0.$$

Therefore,

$$b_z + \delta^2 \nabla_h \beta \cdot \nabla_h H + \delta^2 b_z |\nabla_h H|^2 = 0.$$

Equivalently,

$$\boxed{\left(1 + \delta^2 |\nabla_h H|^2\right) b_z|_{z=-H} + \delta^2 \nabla_h H \cdot \nabla_h \beta = 0.}$$

Now solve the vertical column problem at each horizontal location with prescribed surface value $b_s(x, y)$ and prescribed bottom value $\beta(x, y)$:

$$\varepsilon^2 \partial_z \left(\kappa \partial_z b\right) + \gamma(B - b) = 0, \qquad -H < z < 0,$$

$$b(0) = b_s, \qquad b(-H) = \beta.$$

This vertical solve defines a Dirichlet-to-Neumann map. Namely, it gives the bottom vertical derivative as

$$b_z(-H) = \Lambda[\beta] + r.$$

Here Λ is the columnwise Dirichlet-to-Neumann operator mapping the imposed bottom trace β to the resulting bottom normal derivative, while r is a known contribution coming from the surface value b_s and the interior forcing B.

Substituting

$$b_z(-H) = \Lambda[\beta] + r$$

into the transformed bottom condition gives the reduced horizontal equation for the unknown bottom trace:

$$\boxed{\left(1 + \delta^2 |\nabla_h H|^2\right) \left(\Lambda[\beta] + r\right) + \delta^2 \nabla_h H \cdot \nabla_h \beta = 0.}$$

Once this equation has been solved for $\beta(x, y)$, the full buoyancy field is reconstructed by solving the vertical column problem with

$$b(0) = b_s(x, y), \qquad b(-H) = \beta(x, y).$$

1.1. **Constant diffusivity example.** Assume

$$\kappa = \kappa_0$$

and define

$$\psi = \frac{1}{\varepsilon} \sqrt{\frac{\gamma}{\kappa_0}}.$$

Then the vertical equation becomes

$$b_{zz} - \psi^2 b = -\psi^2 B.$$

For clarity, first consider the homogeneous case $B = 0$. Then

$$b_{zz} - \psi^2 b = 0,$$

with

$$b(0) = b_s, \qquad b(-H) = \beta.$$

The solution is

$$b(z) = \beta \frac{\sinh(-\psi z)}{\sinh(\psi H)} + b_s \frac{\sinh(\psi(z+H))}{\sinh(\psi H)}.$$

Differentiating,

$$b_z(z) = -\psi\beta \frac{\cosh(-\psi z)}{\sinh(\psi H)} + \psi b_s \frac{\cosh(\psi(z+H))}{\sinh(\psi H)}.$$

Evaluating at the bottom $z = -H$ gives

$$b_z(-H) = -\psi \coth(\psi H)\beta + \psi \operatorname{csch}(\psi H) b_s.$$

Therefore, in this case,

$$\Lambda[\beta] = -\psi \coth(\psi H)\beta,$$

and

$$r = \psi \operatorname{csch}(\psi H) b_s.$$

The reduced bottom-trace equation is therefore

$$\left(1 + \delta^2 |\nabla_h H|^2\right) \left[-\psi \coth(\psi H)\beta + \psi \operatorname{csch}(\psi H) b_s\right] + \delta^2 \nabla_h H \cdot \nabla_h \beta = 0.$$

Equivalently,

$$\boxed{\delta^2 \nabla_h H \cdot \nabla_h \beta - \left(1 + \delta^2 |\nabla_h H|^2\right) \psi \coth(\psi H)\beta = -\left(1 + \delta^2 |\nabla_h H|^2\right) \psi \operatorname{csch}(\psi H) b_s.}$$

After solving this horizontal equation for $\beta(x, y)$, the full solution is recovered from

$$\boxed{b(z) = \beta \frac{\sinh(-\psi z)}{\sinh(\psi H)} + b_s \frac{\sinh(\psi(z+H))}{\sinh(\psi H)}.}$$

1.2. **Including an interior buoyancy profile B.** When $B \neq 0$, the vertical equation is

$$b_{zz} - \psi^2 b = -\psi^2 B(z; x, y).$$

The solution may be written as

$$b(z) = b_{\text{hom}}(z) + b_{\text{part}}(z),$$

where

$$b_{\text{hom}}(z) = \beta \frac{\sinh(-\psi z)}{\sinh(\psi H)} + b_s \frac{\sinh(\psi(z+H))}{\sinh(\psi H)}$$

and b_{part} solves

$$b_{\text{part},zz} - \psi^2 b_{\text{part}} = -\psi^2 B,$$

with homogeneous endpoint conditions

$$b_{\text{part}}(0) = 0, \qquad b_{\text{part}}(-H) = 0.$$

Then

$$b_z(-H) = -\psi \coth(\psi H)\beta + \psi \operatorname{csch}(\psi H) b_s + b_{\text{part},z}(-H).$$

Thus,

$$\Lambda[\beta] = -\psi \coth(\psi H)\beta,$$

and

$$r = \psi \operatorname{csch}(\psi H) b_s + b_{\text{part},z}(-H).$$

Using the Dirichlet Green's function $G_D(z,\xi)$ on $[-H,0]$, satisfying

$$\left(\partial_z^2 - \psi^2\right) G_D(z,\xi) = \delta(z-\xi),$$
$$G_D(0,\xi) = 0, \qquad G_D(-H,\xi) = 0,$$

we may write

$$b_{\text{part}}(z) = -\psi^2 \int_{-H}^{0} G_D(z,\xi) B(\xi)\, d\xi.$$

Therefore,

$$b_{\text{part},z}(-H) = -\psi^2 \int_{-H}^{0} \partial_z G_D(-H,\xi) B(\xi)\, d\xi.$$

For constant ψ, the Dirichlet Green's function is

$$G_D(z,\xi) = -\frac{1}{\psi \sinh(\psi H)} \begin{cases} \sinh\left(\psi(z+H)\right)\sinh(-\psi\xi), & z \le \xi, \\ \sinh\left(\psi(\xi+H)\right)\sinh(-\psi z), & z \ge \xi. \end{cases}$$

At the bottom, $z = -H \le \xi$, so

$$\partial_z G_D(-H,\xi) = -\frac{\sinh(-\psi\xi)}{\sinh(\psi H)}.$$

Hence

$$b_{\text{part},z}(-H) = \psi^2 \int_{-H}^{0} \frac{\sinh(-\psi\xi)}{\sinh(\psi H)} B(\xi)\, d\xi.$$

Therefore,

$$\boxed{r = \psi \operatorname{csch}(\psi H) b_s + \psi^2 \int_{-H}^{0} \frac{\sinh(-\psi\xi)}{\sinh(\psi H)} B(\xi)\, d\xi.}$$

The reduced equation with interior forcing is

$$\boxed{\left(1 + \delta^2 |\nabla_h H|^2\right)\left[-\psi \coth(\psi H)\beta + r\right] + \delta^2 \nabla_h H \cdot \nabla_h \beta = 0.}$$

Equivalently,

$$\boxed{\delta^2 \nabla_h H \cdot \nabla_h \beta - \left(1 + \delta^2 |\nabla_h H|^2\right) \psi \coth(\psi H)\beta = -\left(1 + \delta^2 |\nabla_h H|^2\right) r.}$$

1.3. **Interpretation.** The reduction has converted the original sloping-bottom boundary-value problem into two pieces:

(1) A horizontal equation for the bottom trace $\beta(x,y)$:

$$\left(1 + \delta^2 |\nabla_h H|^2\right)\left(\Lambda[\beta] + r\right) + \delta^2 \nabla_h H \cdot \nabla_h \beta = 0.$$

(2) A vertical reconstruction problem:

$$\varepsilon^2 \partial_z(\kappa b_z) + \gamma(B - b) = 0, \qquad b(0) = b_s, \qquad b(-H) = \beta.$$

The only horizontal coupling appears in the bottom-trace equation. In the limit $\delta \to 0$, the equation reduces to

$$\Lambda[\beta] + r = 0,$$

which enforces

$$b_z(-H) = 0$$

column by column.

For the **Dirichlet-to-Neumann reduction / boundary trace approach**, the standard references are from elliptic PDE theory and domain-decomposition theory. The idea is usually called the **Steklov–Poincaré operator**, **Poincaré–Steklov operator**, or **Dirichlet-to-Neumann map**.

2. Solving the reduced bottom-trace equation for β

We consider a closed horizontal domain $D \subset \mathbb{R}^2$, with no lateral inflow boundary prescribed for the reduced bottom-trace problem. The surface boundary condition at $z = 0$ specifies the buoyancy:

$$b(x, y, 0) = b_s(x, y).$$

The bottom trace is

$$\beta(x, y) = b(x, y, -H(x, y)).$$

After applying the Dirichlet-to-Neumann reduction, the sloping-bottom no-flux condition gives a first-order horizontal equation for β. In the constant diffusivity example this equation has the form

$$\delta^2 \nabla_h H \cdot \nabla_h \beta - \left(1 + \delta^2 |\nabla_h H|^2\right) \psi \coth(\psi H) \beta = -\left(1 + \delta^2 |\nabla_h H|^2\right) r,$$

where

$$\psi = \frac{1}{\varepsilon} \sqrt{\frac{\gamma}{\kappa_0}},$$

and $r(x, y)$ is a known forcing contribution determined by the prescribed surface value b_s and the interior buoyancy profile B.

Equivalently, define

$$\mathbf{a}(x, y) = -\delta^2 \nabla_h H(x, y),$$
$$\lambda(x, y) = \left(1 + \delta^2 |\nabla_h H|^2\right) \psi \coth(\psi H),$$

and

$$q(x, y) = \left(1 + \delta^2 |\nabla_h H|^2\right) r(x, y).$$

Then the reduced bottom-trace equation is

$$\boxed{\mathbf{a} \cdot \nabla_h \beta + \lambda \beta = q.}$$

Since

$$\lambda(x, y) > 0,$$

this is a linear advection–reaction equation with positive reaction.

2.1. **Characteristics.** The characteristic curves $\mathbf{X}(s)$ satisfy

$$\frac{d\mathbf{X}}{ds} = \mathbf{a}(\mathbf{X}) = -\delta^2 \nabla_h H(\mathbf{X}).$$

Therefore the characteristics are downhill gradient-flow curves of the topography H. Indeed,

$$\frac{d}{ds} H(\mathbf{X}(s)) = \nabla_h H \cdot \frac{d\mathbf{X}}{ds} = -\delta^2 |\nabla_h H|^2 \leq 0.$$

Thus H is non-increasing along characteristics. Equality holds only at critical points of H, where

$$\nabla_h H = 0.$$

Along a characteristic,

$$\frac{d}{ds}\beta(\mathbf{X}(s)) = \mathbf{a}(\mathbf{X}(s)) \cdot \nabla_h \beta,$$

and the PDE becomes the scalar ODE

$$\boxed{\frac{d\beta}{ds} + \lambda(\mathbf{X}(s))\beta = q(\mathbf{X}(s)).}$$

The solution along a characteristic from s_0 to s is

$$\boxed{\beta(s) = \beta(s_0) \exp\left[-\int_{s_0}^{s} \lambda(\mathbf{X}(\tau))\, d\tau\right] + \int_{s_0}^{s} \exp\left[-\int_{\sigma}^{s} \lambda(\mathbf{X}(\tau))\, d\tau\right] q(\mathbf{X}(\sigma))\, d\sigma.}$$

2.2. **Closed domains and critical points.** In a closed domain, there is no external inflow boundary for the first-order equation. Instead, characteristics originate from local maxima of H, or from critical sets where

$$\nabla_h H = 0.$$

At such points the advective term vanishes and the reduced equation becomes purely algebraic:

$$\lambda\beta = q.$$

Therefore, at a critical point,

$$\boxed{\beta = \frac{q}{\lambda}.}$$

For the present coefficients,

$$q = \left(1 + \delta^2 |\nabla_h H|^2\right) r,$$

and

$$\lambda = \left(1 + \delta^2 |\nabla_h H|^2\right) \psi \coth(\psi H).$$

Thus, at any critical point of H, where $|\nabla_h H| = 0$,

$$\boxed{\beta = \frac{r}{\psi \coth(\psi H)} = \frac{r}{\psi} \tanh(\psi H).}$$

This value supplies the upstream value for characteristics that emanate from local maxima of H. The value of β is then propagated downhill along the gradient-flow curves using the characteristic formula.

2.3. **Topology of the characteristic field.** Because the characteristic velocity is

$$\mathbf{a} = -\delta^2 \nabla_h H,$$

the characteristic curves are gradient-flow curves. If H is smooth and $\nabla_h H$ is finite, then solutions of

$$\frac{d\mathbf{X}}{ds} = -\delta^2 \nabla_h H$$

are unique away from critical points. Therefore, two distinct characteristics cannot cross at a regular point.

However, the characteristic topology may still be complicated:

- characteristics descend from local maxima of H;
- they can approach saddles or local minima;
- separatrices divide regions fed by different maxima;
- different characteristic families may meet at critical points;
- if upstream values differ across separatrices, β may develop sharp jumps or weak discontinuities there.

The positive reaction term $\lambda\beta$ damps sensitivity to upstream data. Differences in upstream values decay approximately like

$$\exp\left[-\int \lambda \, ds\right]$$

along characteristics. However, without horizontal diffusion, the equation does not smooth across characteristics or across separatrices.

2.4. **Numerical solution by upwind discretization.** For computation, it is often preferable to solve

$$\mathbf{a} \cdot \nabla_h \beta + \lambda\beta = q$$

directly using an upwind finite-difference or finite-volume method rather than explicitly tracing all characteristics.

On a Cartesian grid, write

$$\mathbf{a} = (a_x, a_y).$$

Then

$$a_x \partial_x \beta + a_y \partial_y \beta + \lambda\beta = q.$$

Use upwind differences:

$$\partial_x \beta \approx \begin{cases} \dfrac{\beta_{i,j} - \beta_{i-1,j}}{\Delta x}, & a_x > 0, \\ \dfrac{\beta_{i+1,j} - \beta_{i,j}}{\Delta x}, & a_x < 0, \end{cases}$$

and

$$\partial_y \beta \approx \begin{cases} \dfrac{\beta_{i,j} - \beta_{i,j-1}}{\Delta y}, & a_y > 0, \\ \dfrac{\beta_{i,j+1} - \beta_{i,j}}{\Delta y}, & a_y < 0. \end{cases}$$

This produces a sparse linear system

$$A\boldsymbol{\beta} = \mathbf{q}.$$

The reaction coefficient $\lambda > 0$ makes the problem well-conditioned relative to a pure advection problem.

In a closed domain, one does not impose an external inflow condition. Instead, the behavior near critical points of H determines the upstream values through the algebraic balance

$$\beta = \frac{q}{\lambda}.$$

A robust discretization should therefore treat cells where

$$|\nabla_h H| \approx 0$$

as reaction-dominated cells and set, or strongly relax toward,

$$\beta \approx \frac{q}{\lambda}.$$

2.5. **Small-δ expansion.** Since the horizontal advection term is proportional to δ^2, it may be useful to solve the reduced equation perturbatively. The equation is

$$-\delta^2 \nabla_h H \cdot \nabla_h \beta + \lambda \beta = q,$$

where

$$\lambda = \left(1 + \delta^2 |\nabla_h H|^2\right) \psi \coth(\psi H),$$

and

$$q = \left(1 + \delta^2 |\nabla_h H|^2\right) r.$$

Expand

$$\beta = \beta^{(0)} + \delta^2 \beta^{(1)} + \cdots,$$
$$\lambda = \lambda_0 + \delta^2 \lambda_1 + \cdots,$$
$$q = q_0 + \delta^2 q_1 + \cdots.$$

At leading order,

$$\lambda_0 \beta^{(0)} = q_0,$$

where

$$\lambda_0 = \psi \coth(\psi H), \qquad q_0 = r.$$

Thus

$$\boxed{\beta^{(0)} = \frac{r}{\psi \coth(\psi H)} = \frac{r}{\psi} \tanh(\psi H).}$$

At order δ^2,

$$-\nabla_h H \cdot \nabla_h \beta^{(0)} + \lambda_0 \beta^{(1)} + \lambda_1 \beta^{(0)} = q_1.$$

Therefore,

$$\boxed{\beta^{(1)} = \frac{q_1 + \nabla_h H \cdot \nabla_h \beta^{(0)} - \lambda_1 \beta^{(0)}}{\lambda_0}.}$$

This perturbative approach avoids characteristic-tracing issues and is natural when $\delta^2 \ll 1$.

2.6. **Regularization by horizontal diffusion.** The first-order equation

$$\mathbf{a} \cdot \nabla_h \beta + \lambda \beta = q$$

does not smooth across characteristics. If the characteristic topology contains saddles, separatrices, or reconnecting basins, the solution may develop sharp gradients or weak discontinuities.

A more robust model is obtained by retaining a small horizontal diffusion term in the original buoyancy equation. The reduced bottom-trace equation then becomes an advection–diffusion–reaction problem, schematically

$$-\nabla_h \cdot (D_{\text{eff}} \nabla_h \beta) + \mathbf{a} \cdot \nabla_h \beta + \lambda \beta = q.$$

This is elliptic or advection–diffusion–reaction type and no longer requires delicate characteristic boundary conditions. It also smooths across separatrices.

2.7. **Summary.** The bottom-trace equation in a closed domain is

$$\boxed{\mathbf{a} \cdot \nabla_h \beta + \lambda \beta = q, \qquad \mathbf{a} = -\delta^2 \nabla_h H,}$$

with

$$\lambda = \left(1 + \delta^2 |\nabla_h H|^2\right) \psi \coth(\psi H), \qquad q = \left(1 + \delta^2 |\nabla_h H|^2\right) r.$$

It is a linear advection–reaction equation. The characteristics are downhill gradient-flow curves of H:

$$\frac{d\mathbf{X}}{ds} = -\delta^2 \nabla_h H.$$

At critical points of H, the equation reduces to

$$\beta = \frac{q}{\lambda}.$$

In practice, the equation can be solved either by characteristics, by an upwind finite-volume/finite-difference discretization, or perturbatively in δ^2. If separatrices or reconnecting characteristic basins cause singular behavior, horizontal diffusion should be retained, yielding an advection–diffusion–reaction equation for β.